

XDSM Diagram

Control Inputs

$$U = [\delta_E, \eta_{th}]$$

Atmosphere, Air Data and Flight Conditions

$$V = \sqrt{u^2 + w^2}, \alpha = \arctan(w/u), h = -P_D$$

$$\bar{q} = \frac{1}{2} \rho |h| V^2, S$$

Propulsion Force

$$T = \frac{\eta_p \eta_{th} P_{max}}{V}$$

$$T = \text{body axis}$$

$$[F_T; 0; 0]$$

δ_E

Aerodynamic Coefficients

$$C_L, C_D, C_m$$

Aerodynamic Forces

$$F_L^w, F_D^w, M_y^B$$

Coordinate Transformation
Wind Axis $\rightarrow$ Body Axis

$$F_L^B, F_D^B, M_y^B$$

Gravitational Force, Coordinate Transformation

$$F_g^B$$

Total Force & Moments in Body Frame in Vectorial Form

$$F_x, F_y, M_y$$

Coupled & Solved
 $\ddot{u}, \ddot{w}, \ddot{\alpha}$

Longitudinal EOM
 $\dot{X} = [u, w, q, \alpha, P, h]$

Integrate $\int$

Stop

EOM of Cessna-182

① Translational EOM Velocity

→ Newton's 2nd Law = $\vec{F} = \frac{d}{dt} [m \vec{V}]$ // eq. 1

eq. 2 $\vec{F} = m \dot{\vec{V}}$ // fuel burn is neglected

→ $\dot{\vec{V}}$ can be expressed also by Coriolis Equation

$$\dot{\vec{V}} = {}^b\dot{\vec{V}} + \vec{\omega}_{b/i} \times \vec{V} \quad \text{eq. 3}$$

→ Subst. eq. 3 in to eq. 2

$$\vec{F} = m [{}^b\dot{\vec{V}} + \vec{\omega}_{b/i} \times \vec{V}] \quad \text{eq. 4}$$

$${}^b\dot{\vec{V}} = \frac{1}{m} \vec{F} - \vec{\omega}_{b/i} \times \vec{V} \quad \text{eq. 5}$$

→ Expressed in body frame:

$${}^b\dot{\vec{V}}^b = \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \frac{1}{m} \vec{F}^b - \vec{\omega}_{b/i}^b \times \vec{V}^b$$

$\vec{V} = \vec{p} = \vec{r} = \vec{0} \Rightarrow \vec{\omega}_{b/i}^b \times \vec{V}^b = \begin{bmatrix} q w \\ 0 \\ -q u \end{bmatrix}, \vec{F}^b = [F_x, F_y, F_z]$

Result:

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \begin{bmatrix} \frac{F_x}{m} - q w \\ \frac{F_y}{m} \\ \frac{F_z}{m} + q u \end{bmatrix}$$

3 DOF Longitudinal Translational Velocity Dynamic.

② Rotational EOM (Angular Velocity)

→ Newton's 2nd Law Rotation $\vec{M} = \frac{d}{dt} \left[\vec{H} \right]$,

where $\vec{H} = \mathbf{I}_b \vec{\omega}_{b/i}$ → Defined in body frame
Angular momentum Angular vel.
Inertia Matrix

→ In inertial frame measuring $\frac{d}{dt} \left[\vec{H} \right]$ is hard. Change to body frame using Coriolis Eq

$$\vec{M} = \frac{d}{dt} \left[\mathbf{I}_b \vec{\omega}_{b/i} \right] + \vec{\omega}_{b/i} \times \mathbf{I}_b \vec{\omega}_{b/i} \quad \text{if } \mathbf{I}_b \text{ constant}$$

Doing some algebra ~

$$\mathbf{I}_b^{-1} \vec{M} = \dot{\vec{\omega}}_{b/i} + \vec{\omega}_{b/i} \times \mathbf{I}_b \vec{\omega}_{b/i}$$

→ No gyroscopic cross-coupling

□ $p=r=0$ → when we substitute;

Result:

$$\ddot{\theta} = \frac{M_y}{I_{yy}}$$

③ Rotational EOMs (Angular Position)

From Euler kinematic's Equation → $\dot{\vec{\Phi}} = \mathbf{H}(\vec{\Phi}) \vec{\omega}_{b/i}$

$p=r=0$, where $\mathbf{H}(\vec{\Phi}) = \begin{bmatrix} 1 & +\theta \sin \phi & +\theta \cos \phi \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \phi & \cos \phi / \cos \phi \end{bmatrix}$

$\phi = 0$

↓
when we substitute in $\dot{\vec{\Phi}} = \mathbf{H}(\vec{\Phi}) \vec{\omega}_{b/i}$

Result: $\dot{\theta} = \theta$

④ Translational COM (Translational Position)

$$\bar{V}^v = C_{vib}(\Phi) \bar{V}^b \quad // \text{Velocity in Inertial Frame}$$

$$\begin{bmatrix} \dot{\bar{p}}_N \\ \dot{\bar{p}}_E \\ \dot{\bar{p}}_D \end{bmatrix} = C_{vib}(\Phi) \bar{V}^b$$

$\rightarrow \phi = \psi = 0 \rightarrow$ Simplifies the DCM $C_{vib}(\Phi)$

$\rightarrow V=0$ in $\bar{V}^b$

Result:
$$\begin{bmatrix} \dot{\bar{p}}_N \\ \dot{\bar{p}}_D \end{bmatrix} = \begin{bmatrix} u \cos \theta + w \sin \theta \\ -u \sin \theta + w \cos \theta \end{bmatrix}$$

$\Rightarrow$ Combining All Equations

$$\dot{\bar{X}} = \begin{bmatrix} \dot{\bar{p}}_N \\ \dot{\bar{p}}_D \\ \dot{\phi} \\ \dot{\psi} \\ \dot{\theta} \\ \dot{\alpha} \end{bmatrix}$$

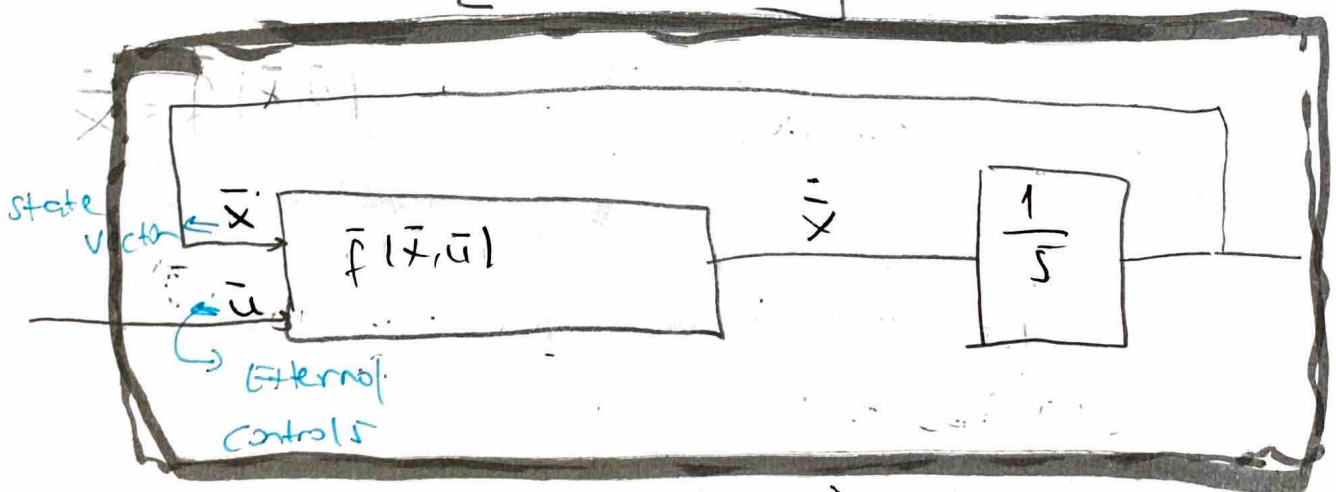
$$= \begin{bmatrix} \frac{F_x^b}{m} - g \sin \theta \\ \frac{F_z^b}{m} + g \cos \theta \\ \frac{M_y^b}{J_y} \\ 0 \\ -u \cos \theta + w \sin \theta \\ -u \sin \theta + w \cos \theta \end{bmatrix}$$

where:

$$\bar{F}^b = \bar{F}_A^b + \bar{F}_T^b + \bar{F}_g^b$$

$$\bar{F}^b = [F_x; F_y; F_z]$$

$$m_y^b = m_A^b$$



Cessna-182 Nonlinear Dynamic System

→ AIC is equipped w/ 230 HP.

→ To convert SI units; $P_{max} = 230 \cdot (745.7) = 171.5 \text{ kW}$

$$u \in [0, 1] \quad P_{shaft} = u P_{max}$$

→ Constant propeller - efficiency model;

$$\text{Propulsive Power, } P_{prop} = \eta_p P_{shaft}$$

$$\text{and also } P_{prop} = T \cdot V$$

$$T = \frac{\eta_p P_{shaft}}{V}$$

② Air, Sea, Atmosphere and Flight Conditions

2.2 Density (ρ) changes w.r.t altitude (h), ISA

→ This technique is valid until troposphere.

$$① T = T_0 - 6.5 \frac{h(m)}{1000}$$

Temperature
model

$$T = T_0 - Lh \rightarrow \text{where } L = \frac{6.5}{1000}$$

$$② \frac{dp}{p} = -\frac{g}{RT} dh \rightarrow \text{How pressure decreases w. altitude}$$

$$\text{substitute } T = T_0 - Lh$$

$$p = p_0 \left(1 - \frac{Lh}{T_0} \right)^{\frac{g_0}{RL}}$$

$$③ \text{Ideal gas law} \Rightarrow \rho = \frac{p}{RT}$$

$$\frac{\rho}{\rho_0} = \frac{p/RT}{p_0/RT_0} = \frac{p}{p_0} \cdot \frac{T_0}{T}$$

$$= \left(\frac{T}{T_0} \right)^{\frac{g_0}{RL}} \left(\frac{T}{T_0} \right)^{-1} \quad // \text{ substitute w/ } \frac{p}{p_0} = \left(\frac{T}{T_0} \right)^{\frac{g_0}{RL}}$$

$$\rho = \left(\frac{T}{T_0} \right)^{\frac{g_0}{RL} - 1}$$

2.5 Gravity Model

Gravity acceleration decreases w.r.t altitude.

$$g(h) = g_0 \left(\frac{R_e}{R_e + h} \right)^2$$

Force Decomposition [Total Forces]

$$\vec{F}^b = \vec{F}_A^b + \vec{F}_T^b + \vec{F}_g^b$$

$$\rightarrow \vec{F}_0^V = \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix}, \quad \boxed{\vec{F}_g^b} = C_{b1V} |\vec{x}| \cdot \vec{F}_g^V = mg \begin{bmatrix} -\sin\theta \\ 0 \\ \cos\theta \end{bmatrix}$$

$$\rightarrow \vec{F}_T^b = \begin{bmatrix} T/V, u_{th} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{\rho_{out} P_{max}}{V} \\ 0 \\ 0 \end{bmatrix}$$

$\rightarrow$ Aerodynamic Forces, $\vec{F}_A^b$

$$\rightarrow C_L = C_{L0} + C_{L\alpha}\alpha + C_{Lq}\hat{q} + C_{L\delta\epsilon}\delta\epsilon + \underbrace{\left(C_{L\dot{\alpha}} \frac{\dot{\alpha}}{2V} \right)}_{L_{coeff}}$$

where $\hat{q} = \frac{\rho \bar{c}}{2V}$ $\rightarrow$ Non dimensionalize

$$C_{L-base} = C_{L0} + C_{L\alpha}\alpha + C_{Lq}\hat{q} + C_{L\delta\epsilon}\delta\epsilon$$

$$L_{base} = q \cdot S \cdot C_{L-base}, \quad \boxed{L_{coeff} = q \cdot S \cdot C_{L\dot{\alpha}} \frac{\dot{\alpha}}{2V}}$$

$$\rightarrow C_m = C_{m0} + C_{m\alpha}\alpha + C_{mq}\hat{q} + C_{m\delta\epsilon}\delta\epsilon + \underbrace{\left(C_{m\dot{\alpha}} \frac{\dot{\alpha}}{2V} \right)}_{M_{coeff}}$$

$$C_{m-base} = C_{m0} + C_{m\alpha}\alpha + C_{mq}\hat{q} + C_{m\delta\epsilon}\delta\epsilon$$

$$M_{y-base} = q \cdot S \cdot \bar{c} \cdot C_{m-base}$$

$$\rightarrow C_D = C_{D0} + C_{D\alpha}\alpha + C_{D\delta\epsilon}\delta\epsilon$$

$$\boxed{D = q S C_D}$$

$$L = L_{\text{base}} + L_{\text{coef}} \dot{\alpha}$$

$$\vec{F}_A^b = C_{b/w} \begin{bmatrix} -D \\ 0 \\ -L_{\text{base}} \end{bmatrix} + \dot{\alpha} C_{b/w} \begin{bmatrix} 0 \\ 0 \\ -L_{\text{coef}} \end{bmatrix}$$

$$d\vec{F}_A^b / d\dot{\alpha}$$

$$\frac{d\vec{F}_A^b}{d\dot{\alpha}} = \underbrace{\begin{bmatrix} \cos\alpha & 0 & -\sin\alpha \\ 0 & 1 & 0 \\ \sin\alpha & 0 & \cos\alpha \end{bmatrix}}_{C_{b/w}(\alpha)} \cdot \begin{bmatrix} 0 \\ 0 \\ -L_{\text{coef}} \end{bmatrix} = \begin{bmatrix} L_{\text{coef}} \sin\alpha \\ 0 \\ -L_{\text{coef}} \cos\alpha \end{bmatrix}$$

Total Force without $\dot{\alpha}$ effect;

$$\vec{F}_{\text{total base}}^b = \vec{F}_{\text{base}}^b + \vec{F}_T^b + \vec{F}_{\text{eff}}^b$$

$$\begin{bmatrix} F_{x, \text{base}} \\ 0 \\ F_{z, \text{base}} \end{bmatrix}$$

$$\begin{bmatrix} F_{x, \text{coef}} \\ 0 \\ F_{z, \text{coef}} \end{bmatrix}$$

$$\text{EOM } x \rightarrow \ddot{x} = \frac{F_{x, \text{base}}}{m} - g + \frac{F_{x, \text{coef}}}{m} \dot{\alpha} = A_1 + B_1 \dot{\alpha}$$

$$z \rightarrow \ddot{z} = \frac{F_{z, \text{base}}}{m} + g + \frac{F_{z, \text{coef}}}{m} \dot{\alpha} = A_2 + B_2 \dot{\alpha}$$

$$\text{Def. of } \dot{\alpha} \Rightarrow \dot{\alpha} = \frac{v\dot{w} - w\dot{v}}{v^2} \xleftarrow{\text{Implicit Form}} -w\ddot{w} + v\ddot{v} - \dot{v}\dot{w} = 0$$

In a Matrix Form

$$\begin{bmatrix} 1 & 0 & -B_1 \\ 0 & 1 & -B_2 \\ -w & v & v^2 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{z} \\ \dot{\alpha} \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \\ 0 \end{bmatrix}$$

→ After found $\bar{z}$

$$F_A^b = \begin{bmatrix} -D \cos \alpha + L \sin \alpha \\ 0 \\ -D \sin \alpha - L \cos \alpha \end{bmatrix}$$

Moment Calculation

$$\rightarrow M_g^b = 0, M_T^b = 0, M_A^b = q S \bar{z} C_m$$

$$M_A^b = M_{yy} = \begin{bmatrix} 0 \\ q S \bar{z} C_m \\ 0 \end{bmatrix}$$

Trimming Via Numerical Optimization

$$\dot{\bar{x}} = \bar{f}(\bar{x}, \bar{u}) \quad (p) \text{ find } \bar{x}_0, \bar{u}_0 \text{ for steady state, straight level flight}$$

$$\rightarrow \bar{z} = \begin{bmatrix} \bar{x} \\ \bar{u} \end{bmatrix}, \dot{\bar{z}} = \bar{f}(\bar{z}), (p) \text{ find } \bar{z}_0 \text{ for ss, } S \propto L \text{ flight}$$

→ we can solve this system using the Equality Constraint

Optimization Penalty Method

Constraints

$$\textcircled{1} \dot{x}_1 = \dot{u} = 0$$

$$\textcircled{2} \dot{x}_2 = \dot{\omega} = 0$$

$$\textcircled{3} \dot{x}_3 = \dot{q} = 0$$

$$\textcircled{4} V_A = V_{\text{target}}$$

$$\textcircled{5} \gamma = 0 \quad (\theta - \alpha)$$

$$\textcircled{6} q = 0$$

$$\Rightarrow V_A - V_{\text{target}} = 0$$

Solve $f(\bar{z}) = 0$, (p) min $f_0 = 0$ [find the feasible set]
 (approx) min $f_0 = f_0 + Q^T H Q$

where $Q = \begin{bmatrix} \bar{x}(1) \\ \bar{x}(2) \\ \bar{x}(3) \\ v_0 - v_{\text{target}} \\ y \\ 1 \end{bmatrix}$

Constraint

$H = \text{matrix of penalty parameters}$
 $= \text{diag}[\alpha_1, \alpha_2, \dots, \alpha_n]$
 $= \begin{bmatrix} \alpha_1 & & 0 \\ & \alpha_2 & \\ 0 & & \ddots \\ & & & \alpha_n \end{bmatrix}_{6 \times 6}$

$\rightarrow$ I chose penalty parameter $\alpha_1 = \underline{1}$.

$$\hat{f}_0 = \alpha_1 \bar{x}_1^2 + \alpha_2 \bar{x}_2^2 + \dots$$